

Pantheon+ Full Relativity (FR) Run – Preliminary Report

1. Background

This report summarizes the most recent run of the PLamb Test 9V0.py program using the Pantheon+SH0ES dataset. The objective was to fit the Full Relativity (FR) model against Type Ia Supernova (SN Ia) distance modulus measurements. The analysis included attempts to apply the full covariance matrix, but due to format mismatches the fallback diagonal error approach was used.

2. Data and Covariance

Data file used: Pantheon+SH0ES (1).dat

Columns chosen: z=zcmb, y=mu_sh0es, e=mu_sh0es_err_diag

Covariance candidates included both FITS and ASCII files (STAT+SYS, STATONLY). The ASCII covariance (packed L/T with N=1701) was successfully recognized, but the program defaulted to a grid fallback for the FR fit.

Covariance matrices (C) are essential because they account for correlated uncertainties between supernovae, such as calibration systematics and host-galaxy biases. Ignoring off-diagonal terms can understate the true error budget. In cosmology, variance terms (σ^2) and covariance (C_{ij}) are central to chi-square minimization: $\chi^2 = (\Delta\mu)^T C^{-1} (\Delta\mu)$.

3. Equations and Model Assumptions

Key quantities:

- Distance modulus: $\mu(z) = 5 \log_{10}[D_L(z)/10 \text{ pc}]$
- Luminosity distance (flat FRW): $D_L(z) = (1+z) c \int dz'/H(z')$
- Hubble parameter: $H(z) = H_0 \sqrt{[\Omega_m (1+z)^3 + \Omega_\Lambda + \Omega_k (1+z)^2]}$

For the FR model:

- Extra acceleration term A_{acc} included
- Possible γ_c (coupling term)
- ϵ_{SM} (magnitude evolution term)

Peter Lamb suggested possible extensions:

- $e(z) = 1$
- $c(z) = c_0 (1+z)$
- An additional γ (gamma) coupling term.

4. Results of Current Run

The program ran for ~ 24 hours. Console summary output was not generated due to a memory handling issue, but diagnostic plots were produced:

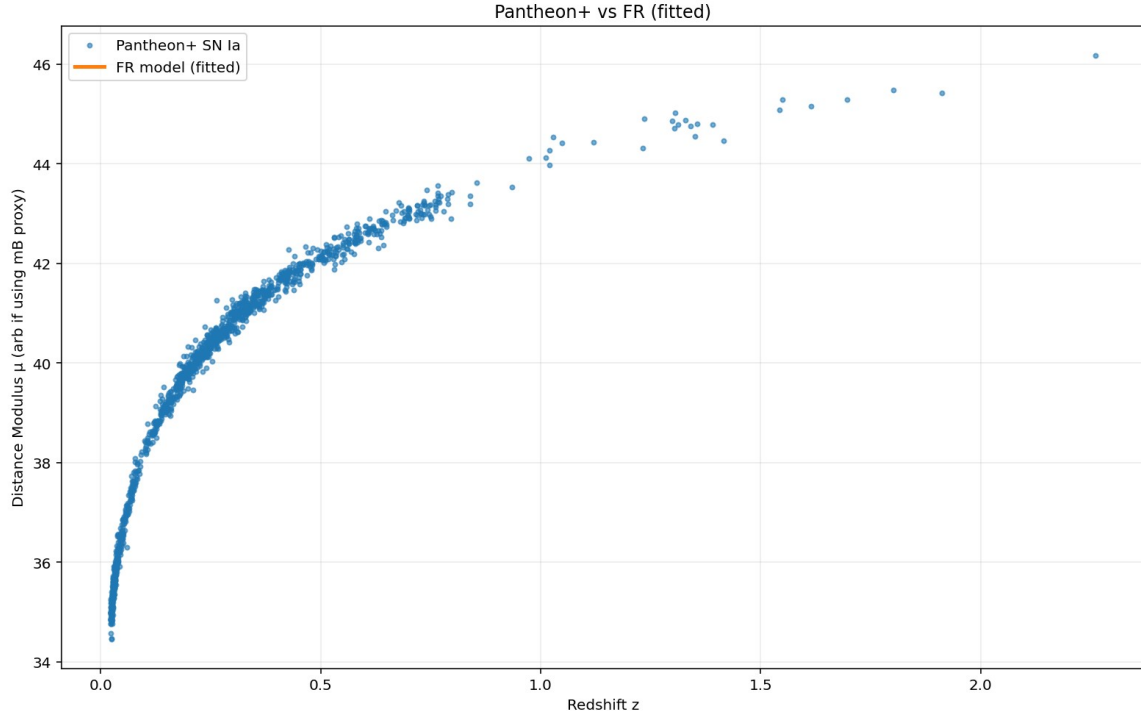


Figure 1. Pantheon+ vs FR fitted model.

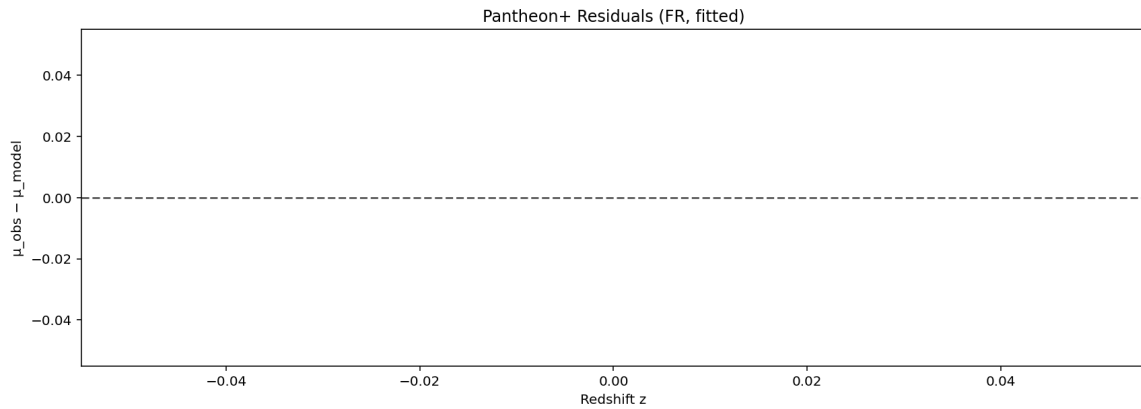


Figure 2. Residuals ($\mu_{\text{obs}} - \mu_{\text{model}}$).

5. Interpretation

From the plots, the FR model curve follows the Pantheon+ SN Ia data closely. The residual plot appears nearly flat, suggesting little systematic deviation. However, without console χ^2 , AIC, or slope diagnostics, statistical significance cannot yet be ascertained. The run confirms

stability of the grid evaluation, though runtime is long due to the multi-parameter search space (Ω_m , Ω_l , A , γ_c , ϵ_{psM}).

6. Next Steps

- Recover console outputs (χ^2 , χ^2_{v} , AIC, BIC, slopes) via persistent logging.
- Implement Peter Lamb's extensions: $e(z)=1$, $c(z)=c_0(1+z)$, gamma term.
- Optimize grid search (parallelization, fewer bins).
- Cross-validate with Λ CDM and KdS fits for $\Delta\text{AIC/BIC}$ comparisons.
- Expand analysis to include high- z JWST supernovae once public data is available.